

Thermodynamics

Basics in thermodynamics

Learning Objectives

Meaning and Significance

Key terms in thermodynamics

Process and properties in thermodynamics

Basics in thermodynamics

SYSTEM $\longrightarrow$ Part of the Universe under study

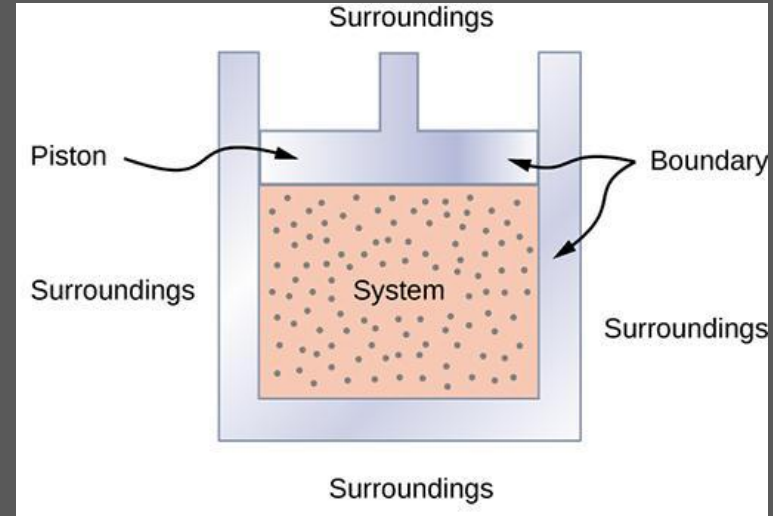
$\downarrow$
All the observation are made on it

SURROUNDINGS

It is remaining part of universe
Can or cannot interact with the system

BOUNDARY

Separation between system and surroundings



$$\text{System} + \text{Surroundings} = \text{Universe}$$

Types Of System

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graph TD; A[Types Of System] --> B[Open System]; A --> C[Closed System]; A --> D[Isolated System];
```

Open System

Closed System

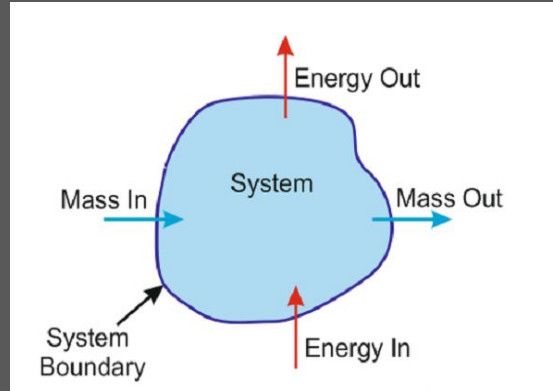
Isolated System

Basics in thermodynamics

Types Of System

Open System

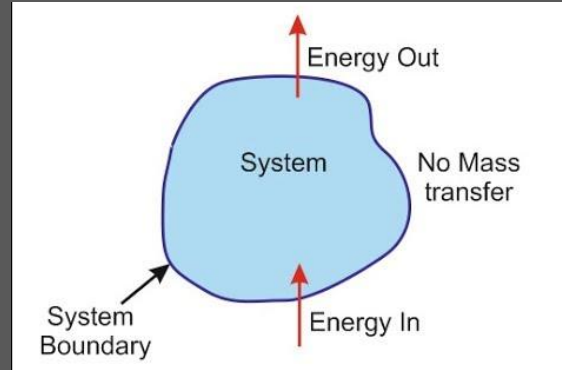
- Exchange of Matter & energy



Types Of System

Closed System

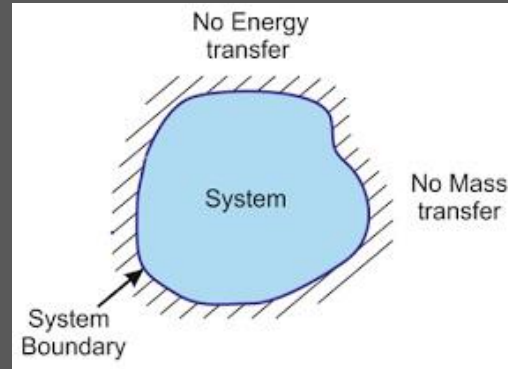
- Only Exchange of energy



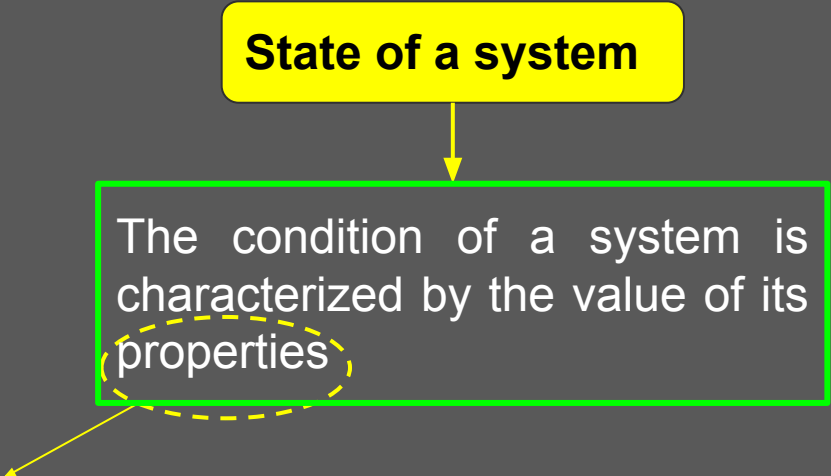
Types Of System

Closed System

- No exchange of matter and energy



State of a system



The condition of a system is characterized by the value of its properties

Properties can be Temperature, pressure, volume and composition of system.

For example state of a system of gas in beaker is defined by its pressure value, temperature, volume and value of moles.

Types of Process

Isothermal

$$\Delta T = 0$$

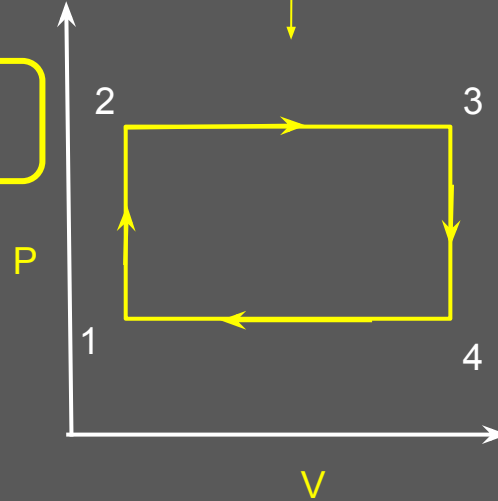
Isochoric

$$\Delta V = 0$$

Isobaric

$$\Delta P = 0$$

Cyclic



Adiabatic

$$\Delta q = 0$$

Types Of Properties

```
graph LR; A[Types Of Properties] --> B[Intensive]; A --> C[Extensive]; B --> D([Properties which do not depend on quantity of matter  
e.g: T, P]); C --> E([Properties which depend on quantity of matter  
e.g: volume, mass])
```

Intensive

Properties which do not depend
on quantity of matter
e.g: T, P

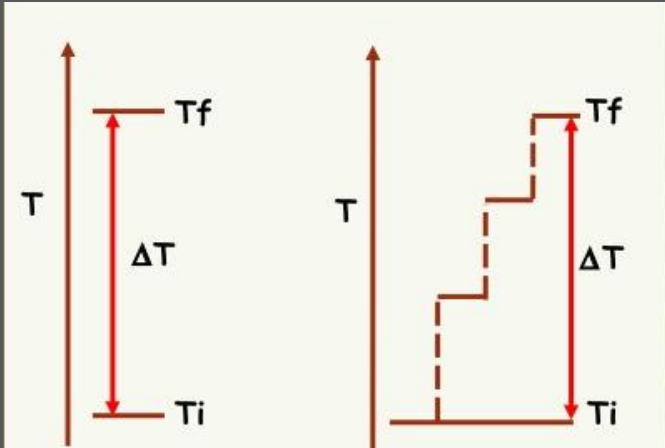
Extensive

Properties which depend on
quantity of matter
e.g: volume, mass

TYPES OF FUNCTIONS

State function: Depends only on **current thermodynamic state** and not on “**how**” the process is obtained.

For Example Changing temperature in single step vs multi step

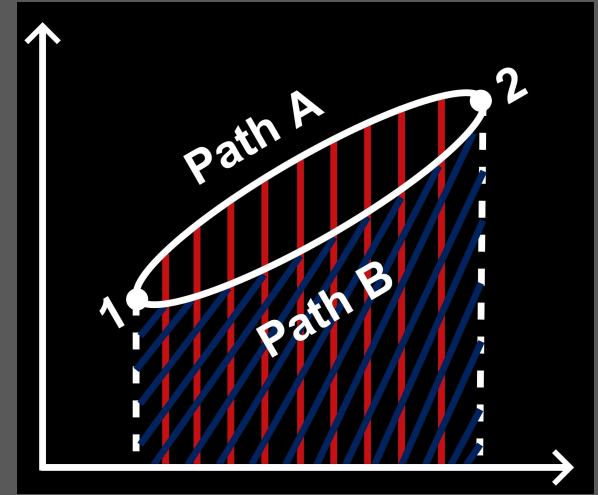
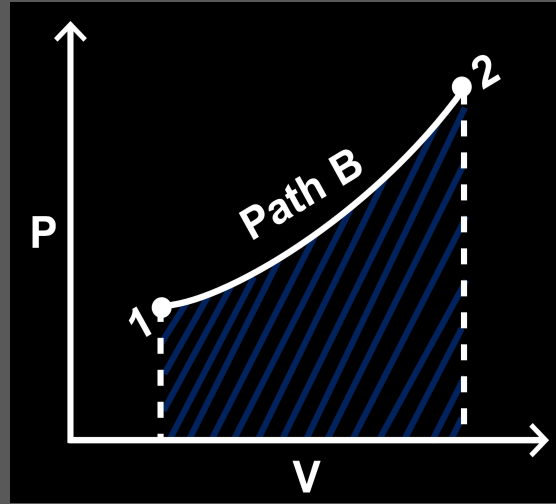
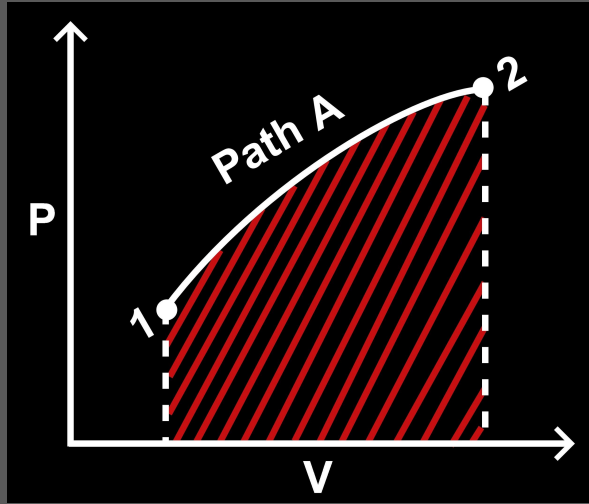


ΔT is same

Other examples are internal energy, enthalpy, volume, pressure etc.

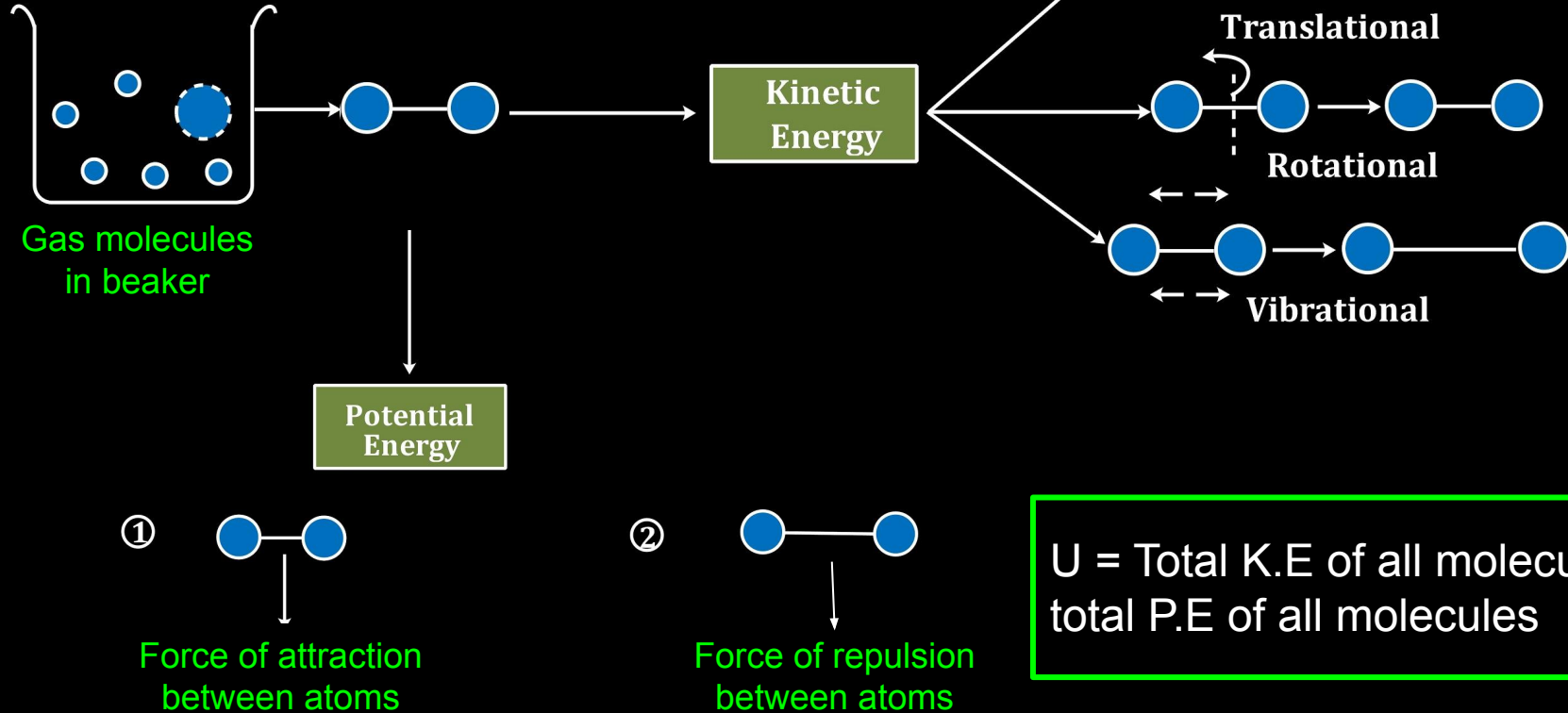
PATH FUNCTION:

Depends on the path of the process, focuses on “how” the final state is attained



Examples - work, heat

INTERNAL ENERGY :



$$U = \text{Total K.E of all molecules} + \text{total P.E of all molecules}$$

Total U can't be measured
only ΔU is measured

S.I unit = Joules (J)

Represented by U

Properties of
Internal energy

State function

Extensive Property

Dependent only on the
change in temperature of
system
 $T \uparrow$ Then Total U $\uparrow$

Internal energy (U) can be changed by

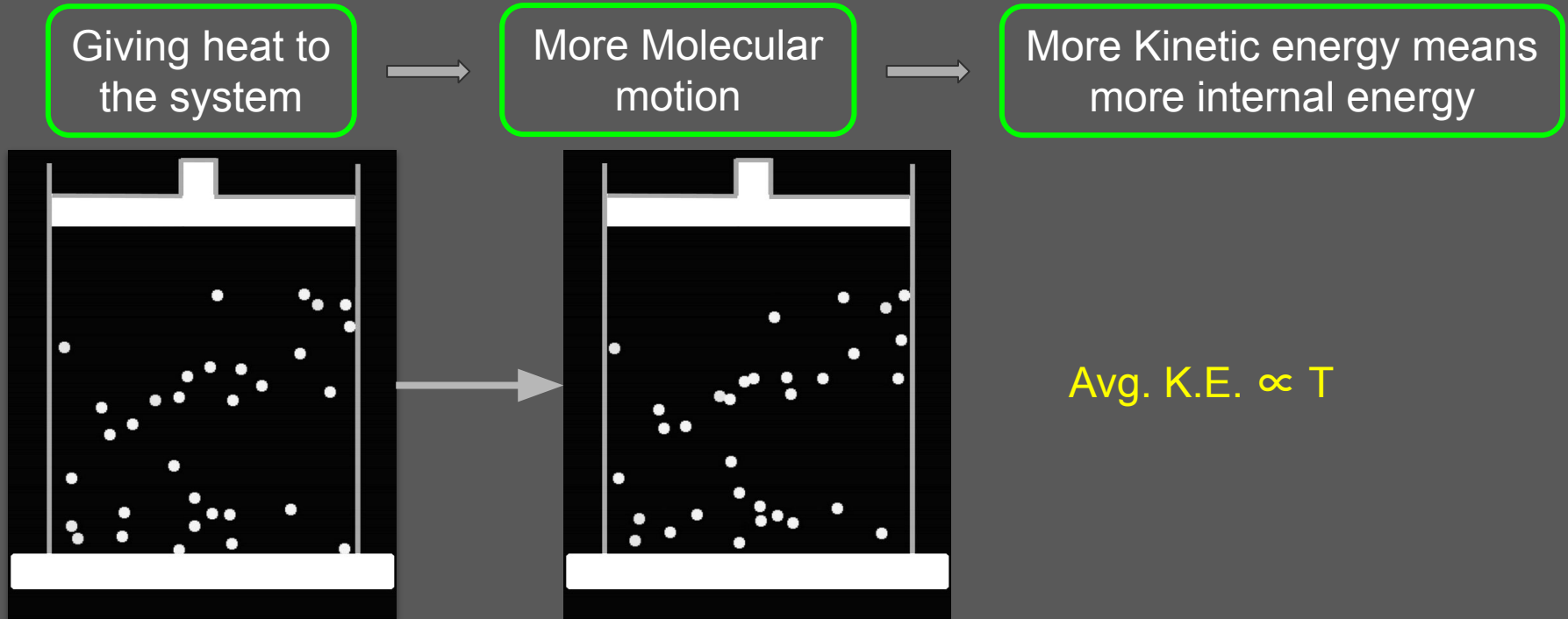
```
graph TD; A[Internal energy (U) can be changed by] --> B[Heat exchange between system and surrounding]; A --> C[Work done on or by the system]; A --> D[Matter enters or leaves the system];
```

Heat exchange
between system
and surrounding

Work done on or
by the system

Matter enters or
leaves the system

a) Heat exchange between system and surrounding:



b) **Work is done on and by the system:** We are doing 1 kJ of work in two ways:

Work can be done in two ways

Work done by the system is “Negative”

Work done on the system is “Positive”

Conclusion : U is a state function

c) Matter enters or leaves the system

Matter transfer
into the system



More number of molecules thus
leading to total increase in internal
energy

Matter transfer
from the system



Less number of molecules thus
leading to total decrease in internal
energy

First Law of thermodynamics

Energy can neither be created nor be destroyed, it can be converted from one form into another.

$$\Delta U = q + w$$

Where q and w are path functions and ΔU is a state function.

If $q=0$, $w=0$ then $\Delta U = 0$

Mechanical work

Energy exchange between system and surrounding against a constant external pressure leading to volume change of system

WORK

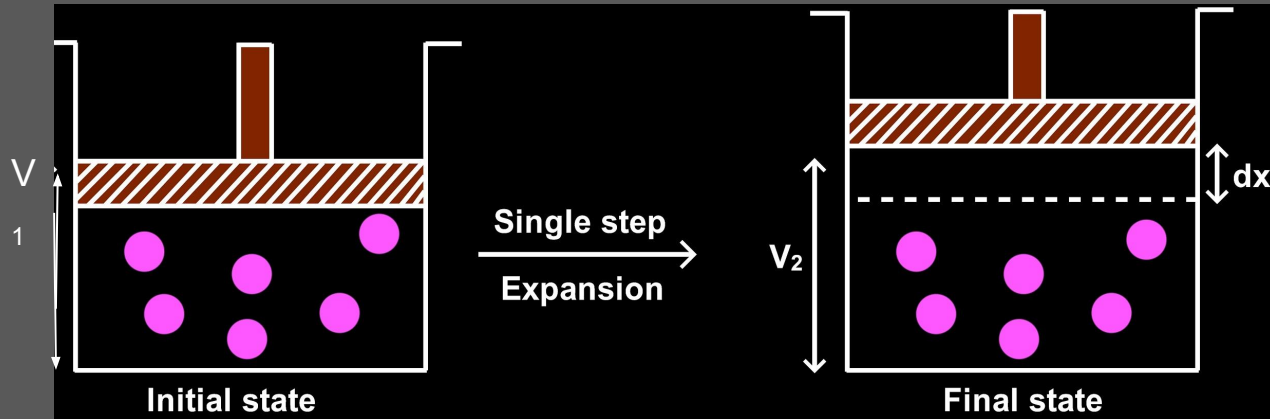
Expansion

$P_{\text{int}} > P_{\text{ext}}$ Gas will expand
against external pressure

Compression

$P_{\text{int}} < P_{\text{ext}}$ Gas will compress
due to external pressure

Derivation of expansion of work



$$w = F \cdot dx$$

$$w = -P_{\text{ex}} A dx$$

$$= -P_{\text{ex}} \cdot (\Delta V)$$

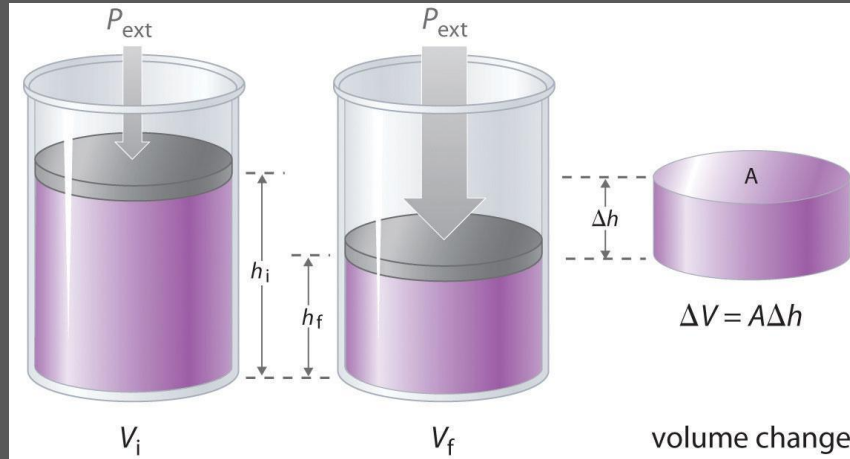
$$w = -P_{\text{ex}} \cdot (V_2 - V_1)$$

Sign Convention of work:

Compression : $V_2 < V_1$

$$\Rightarrow (V_2 - V_1) = \text{negative}, w = -P_{\text{ex}} \cdot (V_2 - V_1) = \text{positive}.$$

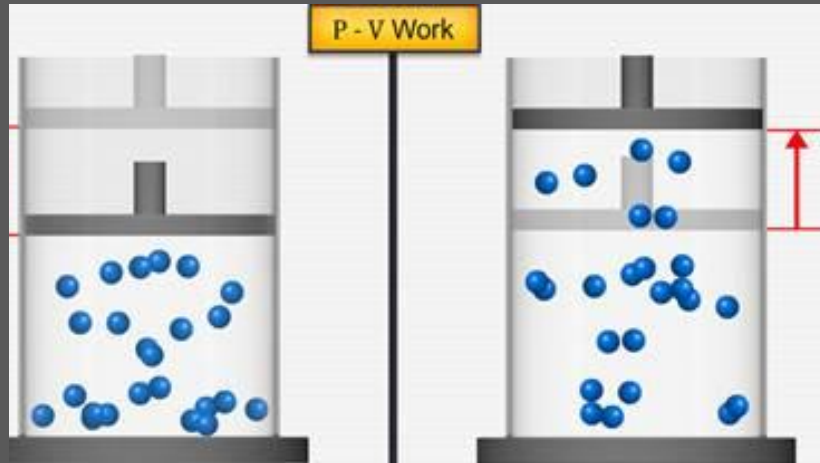
So, Work done **on** the system will always be **positive**



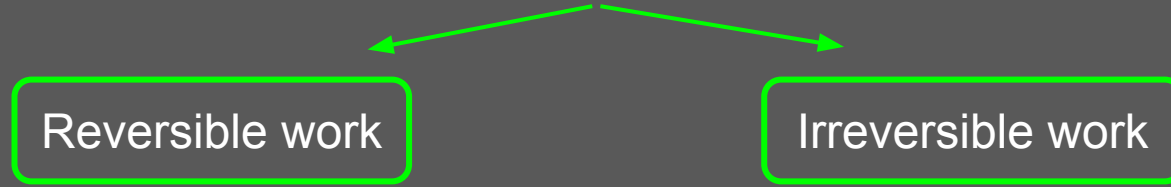
Expansion : $V_2 > V_1$

$\Rightarrow (V_2 - V_1) = \text{positive}$, $w = -P_{\text{ex}} \cdot (V_2 - V_1) = \text{negative}$.

So, “ Work done **by** the system will always be **negative** ”

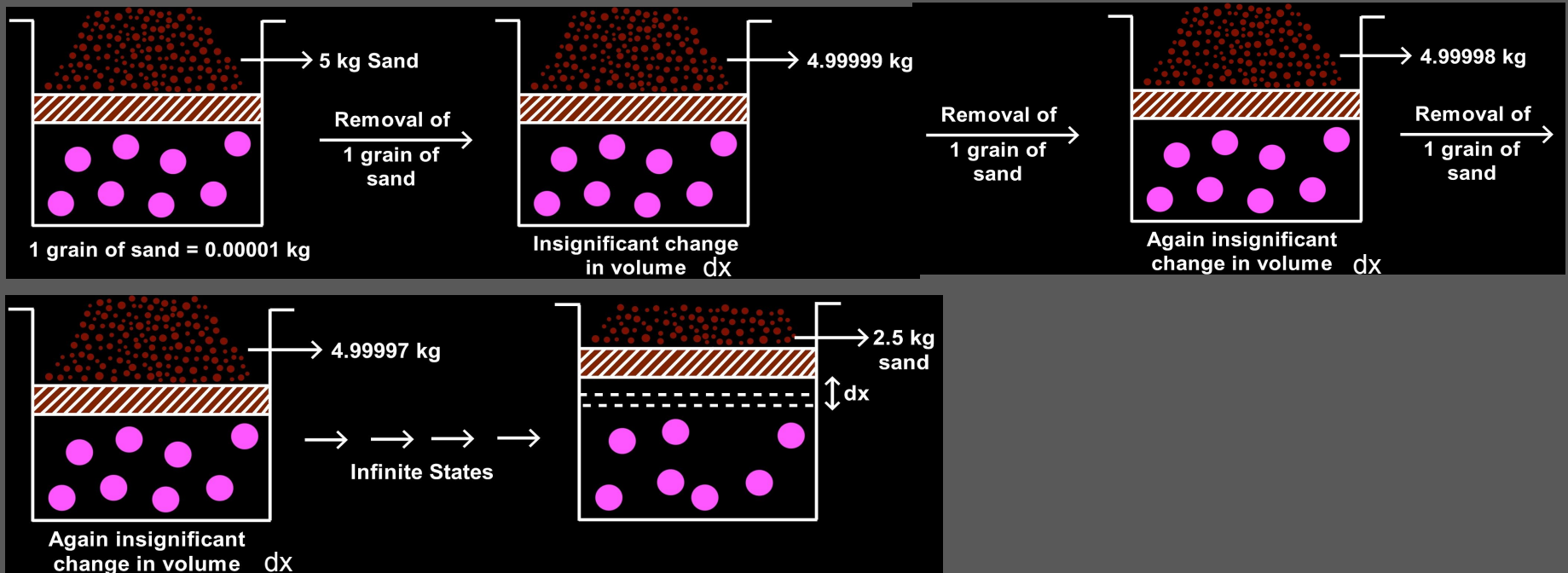


Types of work



Reversible work

Take a beaker with gas molecules and 5 kg sand is used as a weight to show reversible process



$$P_{\text{int}} \approx P_{\text{ex}}$$

Driving force and
opposing force differ
infinitesimally

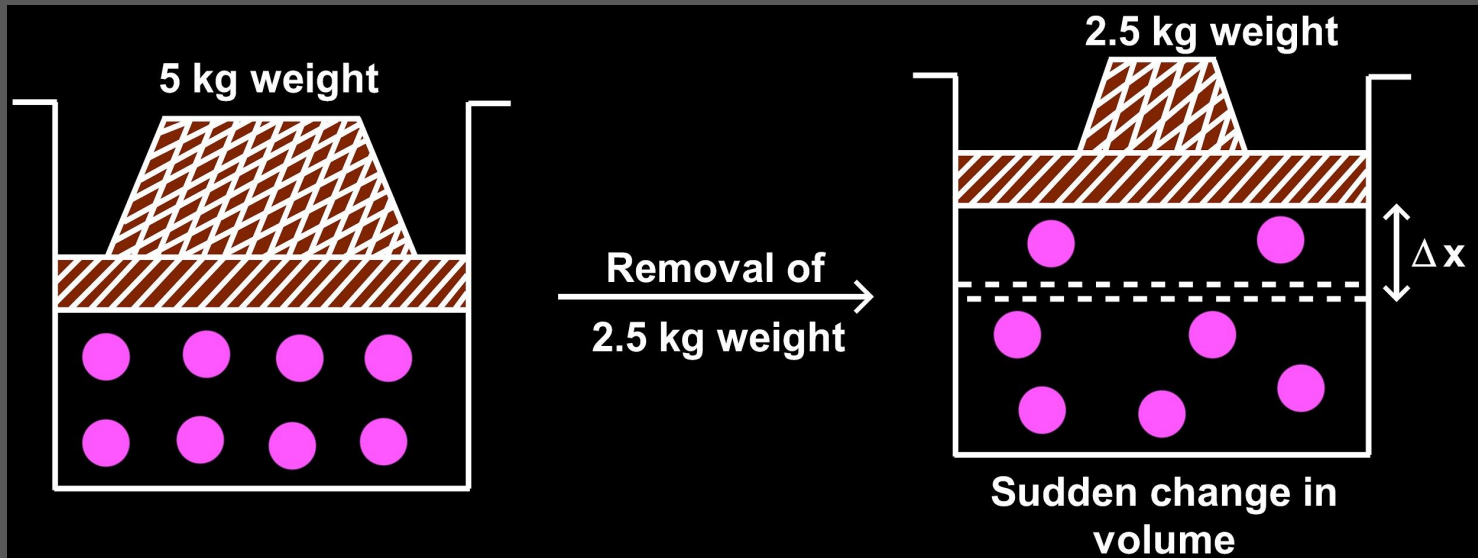
Reversible work

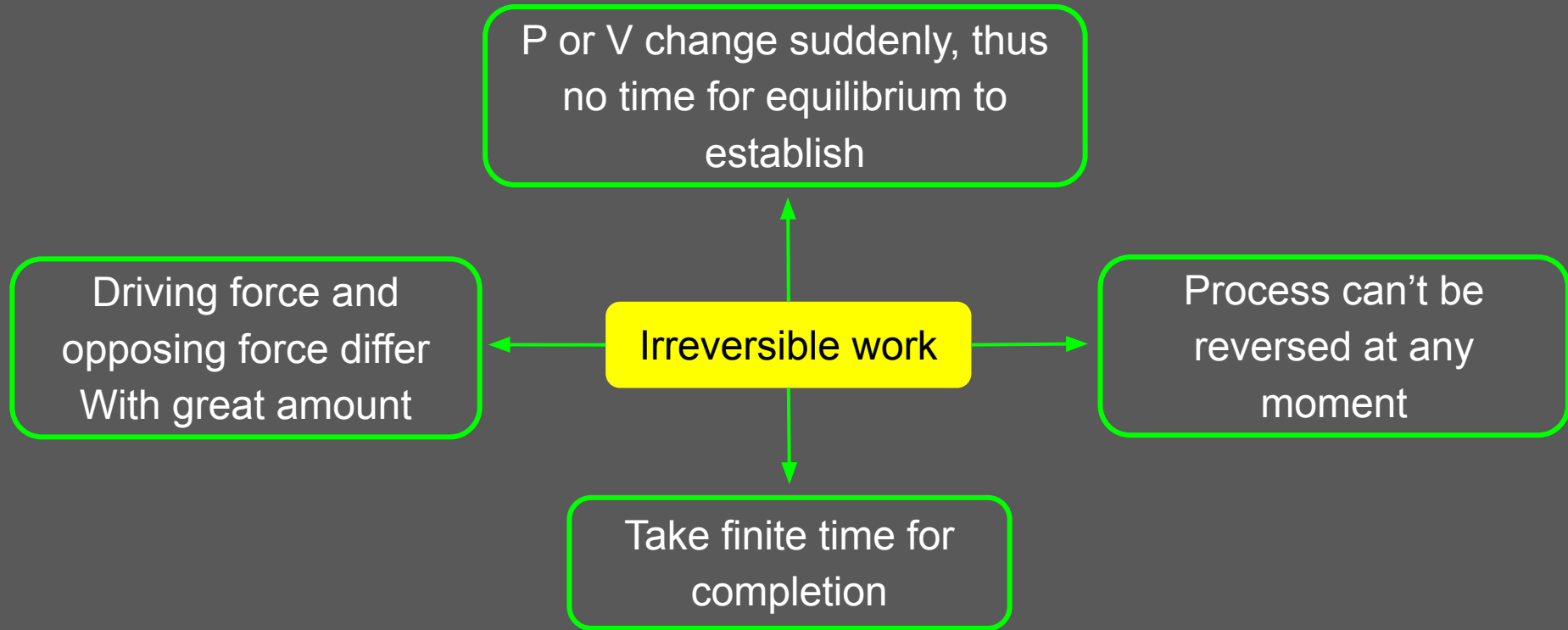
Process can be
reversed at any
moment

Take infinite time for
completion

Irreversible work

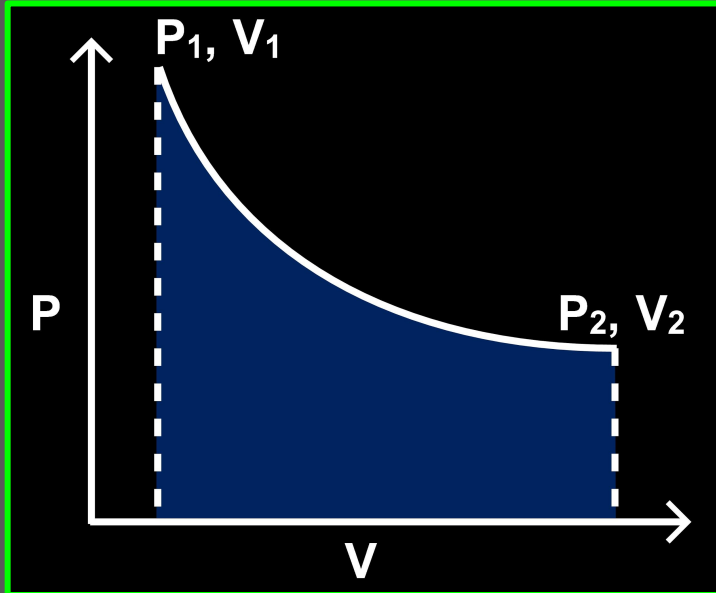
Take a beaker with gas molecules and 5 kg is used as a weight to show irreversible process. Sudden removal of weight caused sudden change in volume



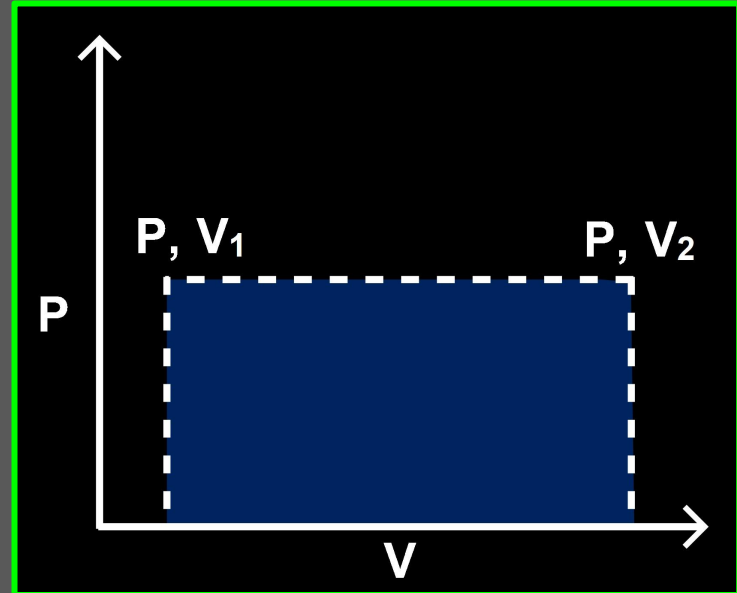


P-V graph for types of expansion work

Reversible work



Irreversible work



Isothermal Reversible work : Let there be ideal gas in beaker with piston

$$w = \int_{V_1}^{V_2} -P_{ex} \cdot dV$$
$$P_{ex} \approx P_{int}$$

Ideal gas Eq. $PV = nRT$

$$P_{int} = \frac{nRT}{V} \quad \dots (1)$$

$$\int_{V_1}^{V_2} -P_{int} \cdot dV \quad \dots (2)$$

Putting (1) in (2)

$$- \int_{V_1}^{V_2} \frac{nRT}{V} \cdot dV$$

T is constant as it isothermal process

$$\Rightarrow -nRT \int_{V_1}^{V_2} \frac{dV}{V} \quad \dots (3)$$

Integrating eq (3)

$$\Rightarrow -nRT [\ln V]_{V_1}^{V_2}$$

$$\Rightarrow -nRT [\ln V_2 - \ln V_1]$$

$$\Rightarrow -nRT \ln \frac{V_2}{V_1} \quad \left[\ln a - \ln b = \ln \frac{a}{b} \right]$$

$$\boxed{w = -nRT \ln \frac{V_2}{V_1}} \quad \dots (4)$$

Reversible work

Some noteworthy points

For isothermal process

$$PV = \text{constant}$$

$$\Rightarrow P_1 V_1 = P_2 V_2$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{V_2}{V_1}$$

$$w = -nRT \ln \frac{V_2}{V_1} \quad \dots (4)$$

Putting (5) in (4)

$$w = -nRT \ln \frac{P_1}{P_2} = -nRT \ln \frac{V_2}{V_1}$$

Q. Two moles of an ideal gas expands reversibly at 25°C until from 2 to 10 litres.
How much work is done in expansion?

Sol. Given - No. of moles, $n = 2$

$$T = 25 + 273 = 298 \text{ K}$$

$$V_1 = 2 \text{ L}, V_2 = 10 \text{ L}$$

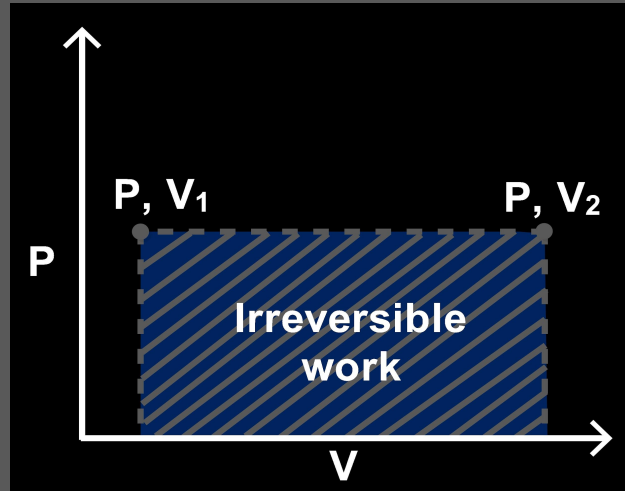
We know isothermal reversible expansion, $w_{\text{rev}} =$

$$\begin{aligned} &= -nRT \ln \frac{V_1}{V_2} \\ &= -2 \times 8.314 \times 298 \times 2.303 \log \frac{10}{2} \\ &= -7976.43 \text{ J} = -7.976 \text{ kJ} \end{aligned}$$

Irreversible work

$$W_{\text{irr}} = -P_{\text{ex}} \cdot \Delta V$$

$$W_{\text{irr}} = -P_{\text{ex}} \cdot (V_2 - V_1)$$



It can also be found out by calculating the area under the curve for P-V graph as it takes place in finite steps.

Free Expansion

$$\text{Vacuum} = P_{\text{ext}} = 0$$

$$W_{\text{free}} = -P_{\text{ex}} \cdot \Delta V$$

$$W_{\text{free}} = 0$$

As $w = 0$,

We know, $\Delta U = q + w$

$$\Delta U = q$$

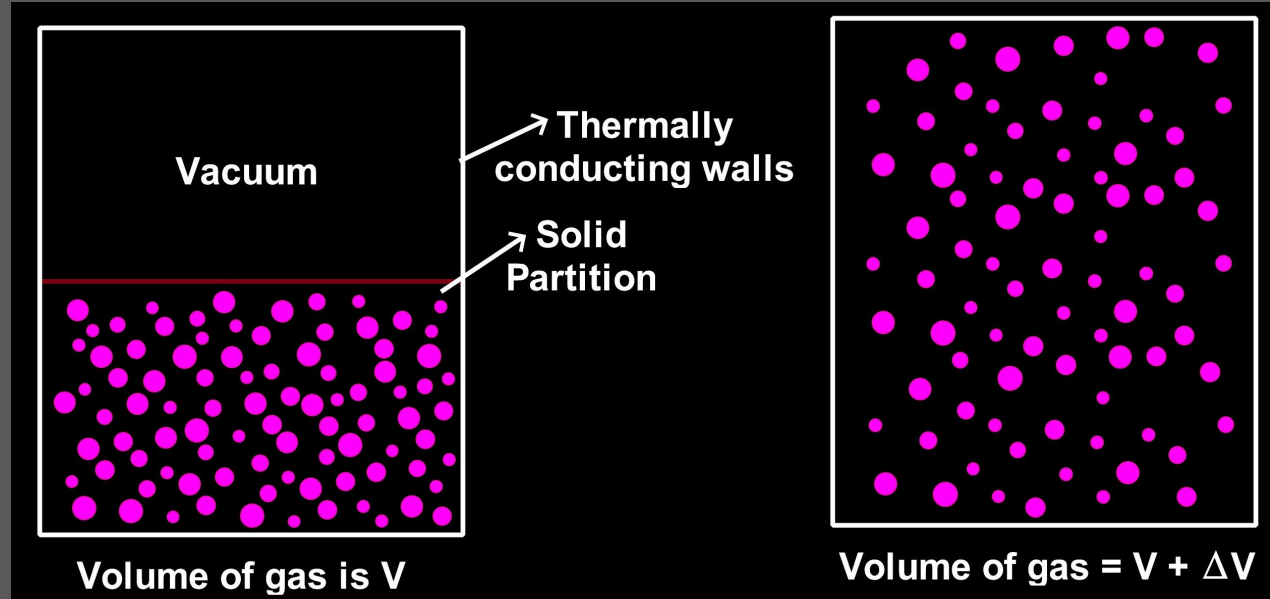


Diagram showing expansion of gas under vacuum

Adiabatic expansion of an ideal gas

$$q = 0$$

According to first law

$$\Delta U = q + w$$

$$\Rightarrow \Delta U = w$$

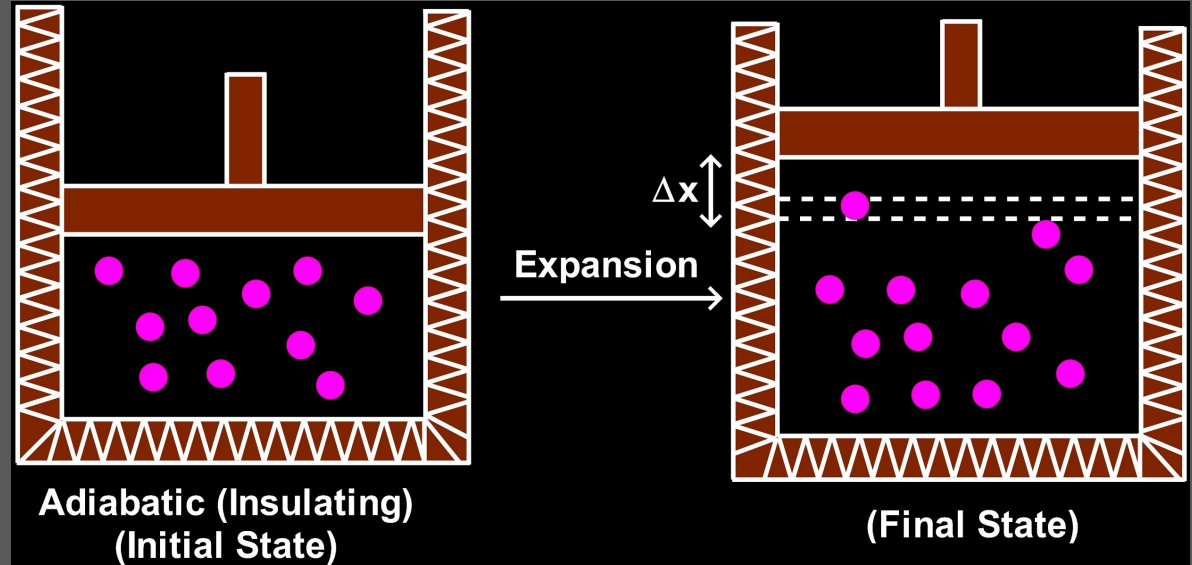


Diagram showing adiabatic expansion of gas

Isothermal Expansion Of an Ideal gas

T is constant implies $\Delta U = 0$ thus, $\Delta U = w+q$ implies $w = -q$

Reversible

Irreversible

Free expansion

$$w_{\text{rev}} = -nRT \ln (V_2/V_1)$$

$$w_{\text{irr}} = -P_{\text{ex}} \cdot \Delta V$$

$$w_{\text{free}} = 0$$

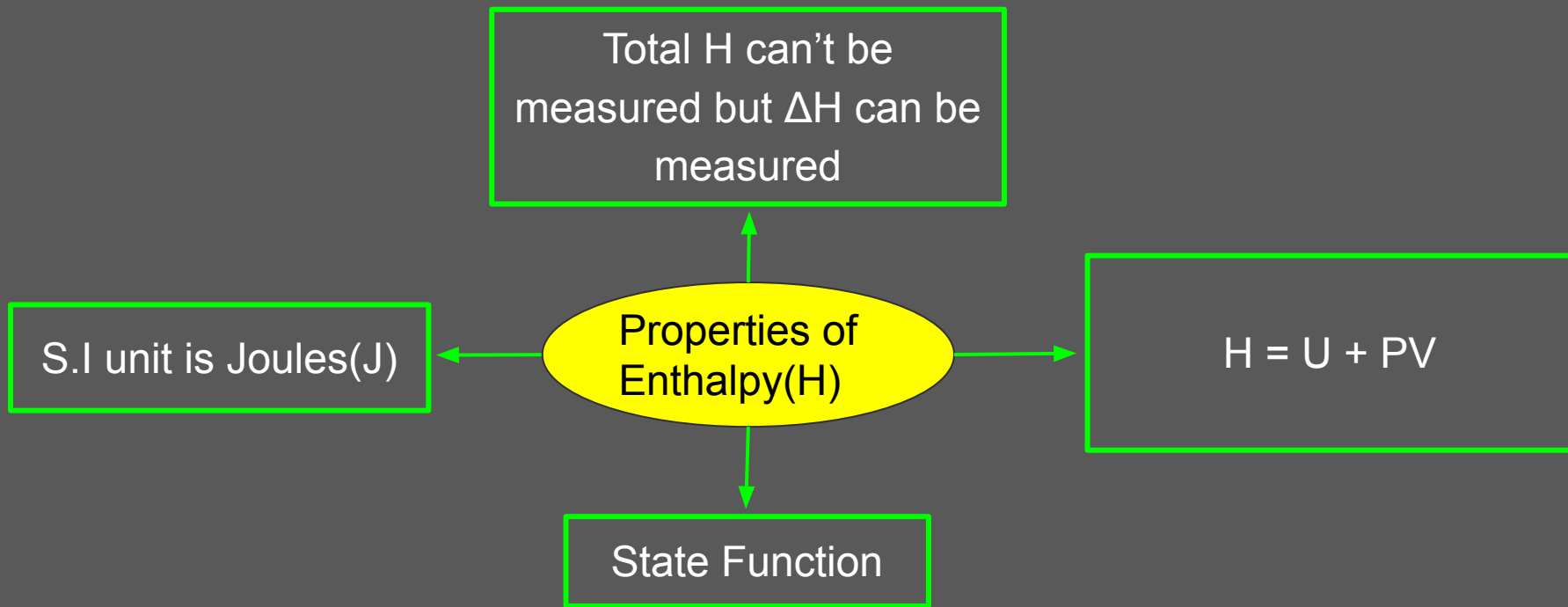
Enthalpy

Total heat content of the system

Heat measured at constant pressure

$$H = U + PV$$

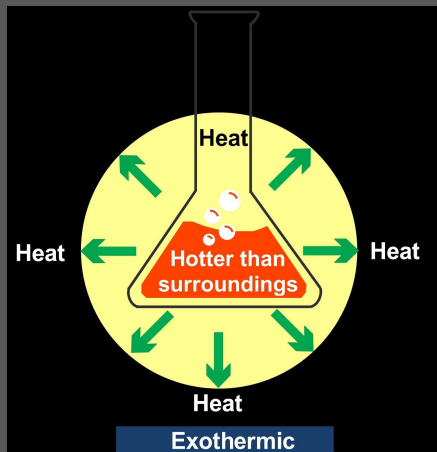
$$\Delta H = \Delta U + P\Delta V$$



Sign convention of ΔH

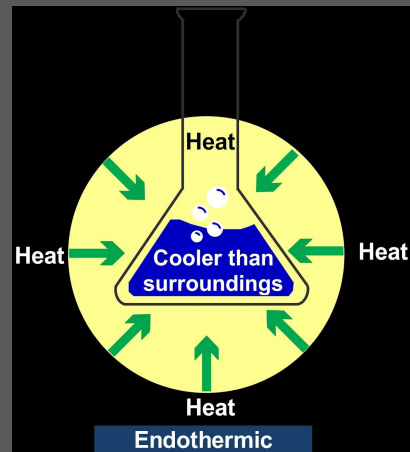
Exothermic Reaction

$\Delta H = \text{negative}$



Endothermic Reaction

$\Delta H = \text{positive}$

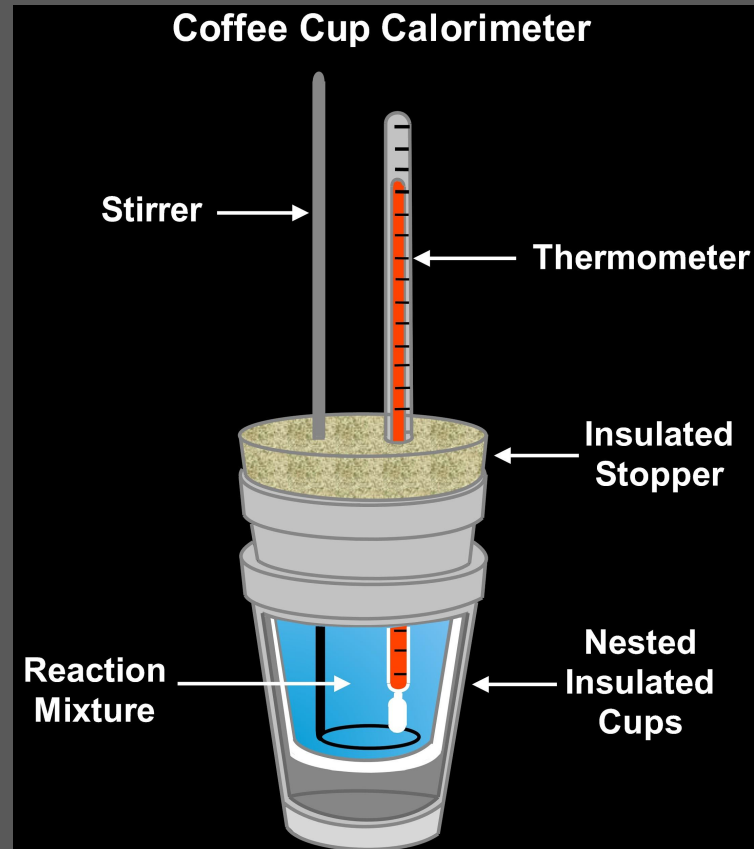


Measurement of ΔH & ΔU

Calorimetry

ΔH measurement

- Measured at **constant pressure**
- **Insulated** cup - prevent exchange of heat
- Thermometer - to measure temperature



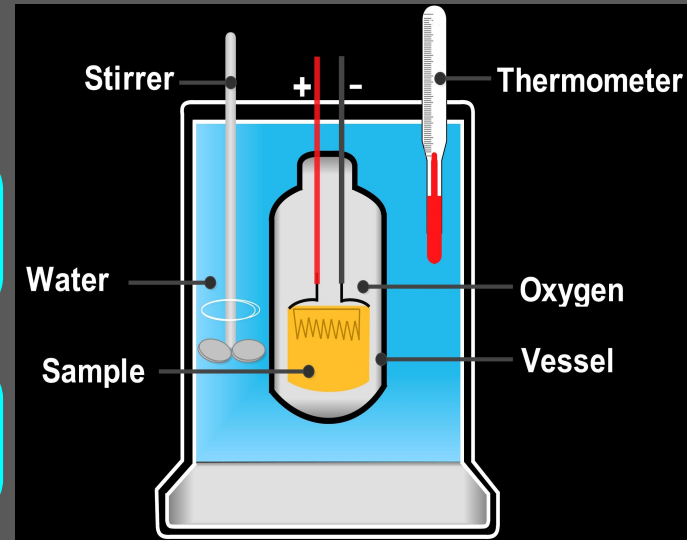
ΔU measurement

- Heat exchanged at **constant volume**
- Measured in **Bomb Calorimeter**

Heat evolved during burning of sample is transferred to water and temperature is measured

Steel vessel in water bath - to transfer heat to water bath as steel vessel is conducting

No volume change as it is sealed, $\Delta V = 0$, thus $w = 0$
 $\Rightarrow \Delta U = q_v$



Heat Capacity

It is the heat required to raise temperature of system by 1 degree celsius

S.I unit is J/K

Extensive Property

Properties of Heat
Capacity (C)

More C means more
heat required to raise
temperature

$$q \propto \Delta T$$
$$q = C \Delta T$$

Types of Heat Capacity

Molar Heat Capacity C_M

Heat required to raise temperature of 1 mole of a system by 1 degree celsius

Intensive property

S.I unit
=J/K-mol

$$C_M = C/n$$
$$q = n C_M \Delta T$$

Specific heat capacity C_S

Heat required to raise temperature of 1 kilogram of a system by 1 degree celsius

Intensive property

S.I unit
=J/K-Kg

$$C_S = C/m$$
$$q = m C_S \Delta T$$



Relationship between C_p & C_v

C_p = Molar Heat Capacity at constant pressure

C_v = Molar Heat Capacity at constant volume

Relationship between C_p & C_v

$$q_v = nC_v \Delta T = \Delta U$$

$$q_p = nC_p \Delta T = \Delta H$$

For ideal gas, $\Delta H = \Delta U + \Delta (PV)$

$$\Rightarrow \Delta H = \Delta U + \Delta (RT) = \Delta U + R\Delta T$$

$$PV = RT \text{ (for 1 mole)}$$

$$C_p \Delta T = C_v \Delta T + R\Delta T$$

$$= \Delta T (C_v + R)$$

$$\Rightarrow C_p - C_v = R$$

Enthalpy of reaction

Energy change accompanying a reaction is enthalpy of reaction

For a reaction

Reactants $\longrightarrow$ Products



$$\Delta_r H = (\text{sum of enthalpies of products}) - (\text{sum of enthalpies of reactants})$$

$$= (c H_C + d H_D) - (a H_A + b H_B)$$

Q. Enthalpy of formation of CO(g), CO₂ (g), N₂O(g), N₂O₄ (g) are -110, -393, 81, 9.7 kJ/mol. Find $\Delta_r H$ for Reaction- N₂O₄ (g) + 3CO(g) $\longrightarrow$ N₂O(g) + 3CO₂ (g)

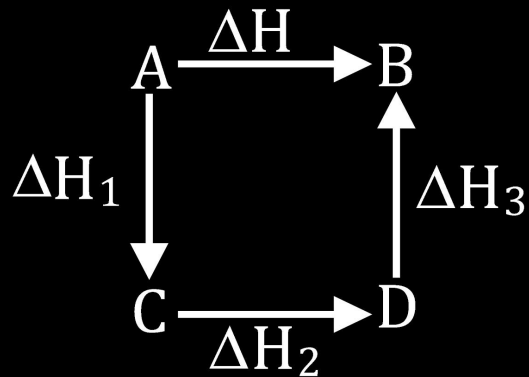
$$\Delta_r H^\circ = \sum \Delta_f H^\circ_{\text{products}} - \sum \Delta_f H^\circ_{\text{reactants}}$$

$$\Delta_r H^\circ = \Delta_f H^\circ(\text{N}_2\text{O}) + 3 \Delta_f H^\circ(\text{CO}_2) - 3 \Delta_f H^\circ(\text{CO}) - \Delta_f H^\circ(\text{N}_2\text{O}_4)$$

$$\Delta_r H = [(-393 \times 3) + 81 - (3 \times (-110)) - 9.7]$$

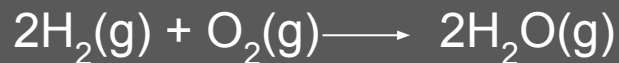
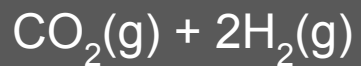
$$= -777.7 \text{ kJ/mol}$$

It states that regardless of the multiple stages or steps of a reaction, the total enthalpy change for the reaction is the sum of all changes.

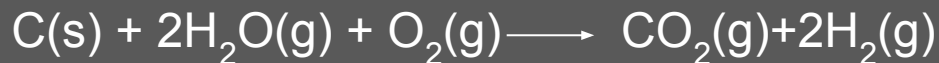


According to Hess's law,

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$



$$\Delta_r H^0 = -483.6 \text{ kJ} \quad \dots \text{ (i)}$$



$$\Delta_r H^0 = -393.5 \text{ kJ}$$



$$\Delta_r H^0 = 90.1 \text{ kJ} \quad \dots \text{ (ii)}$$

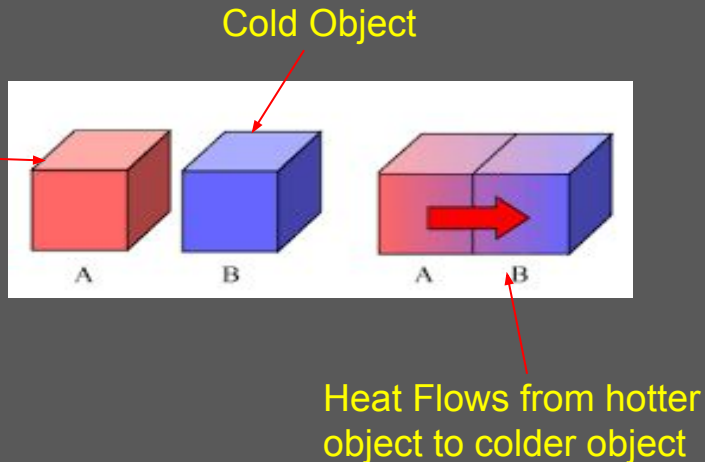
Adding two eq. **Total Enthalpy change = -393.5 kJ**

Spontaneity

Spontaneous Process

Process which **does not need** help of any **external agency** to occur.

Example :



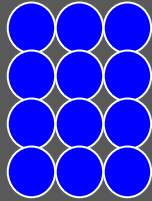
Water flows down the hill

Spontaneous reaction can not be reversed without the help of external agency

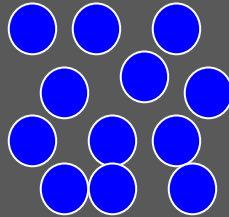
Entropy (S)



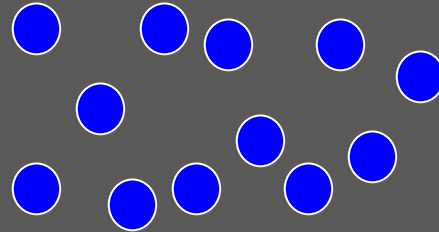
The measure of degree of **disorderness or randomness** in a system.



More ordered
Less entropy



Medium Entropy



Less ordered
More Entropy

State Function

Entropy

$$\Delta S_{\text{solid}} < \Delta S_{\text{liquid}} < \Delta S_{\text{gas}}$$

Extensive Property

$$\Delta S^{\circ}_{\text{reaction}} = \sum \Delta S^{\circ}_{\text{product}} - \sum \Delta S^{\circ}_{\text{reactant}}$$

Q. Predict in which of the following entropy increases/decreases:

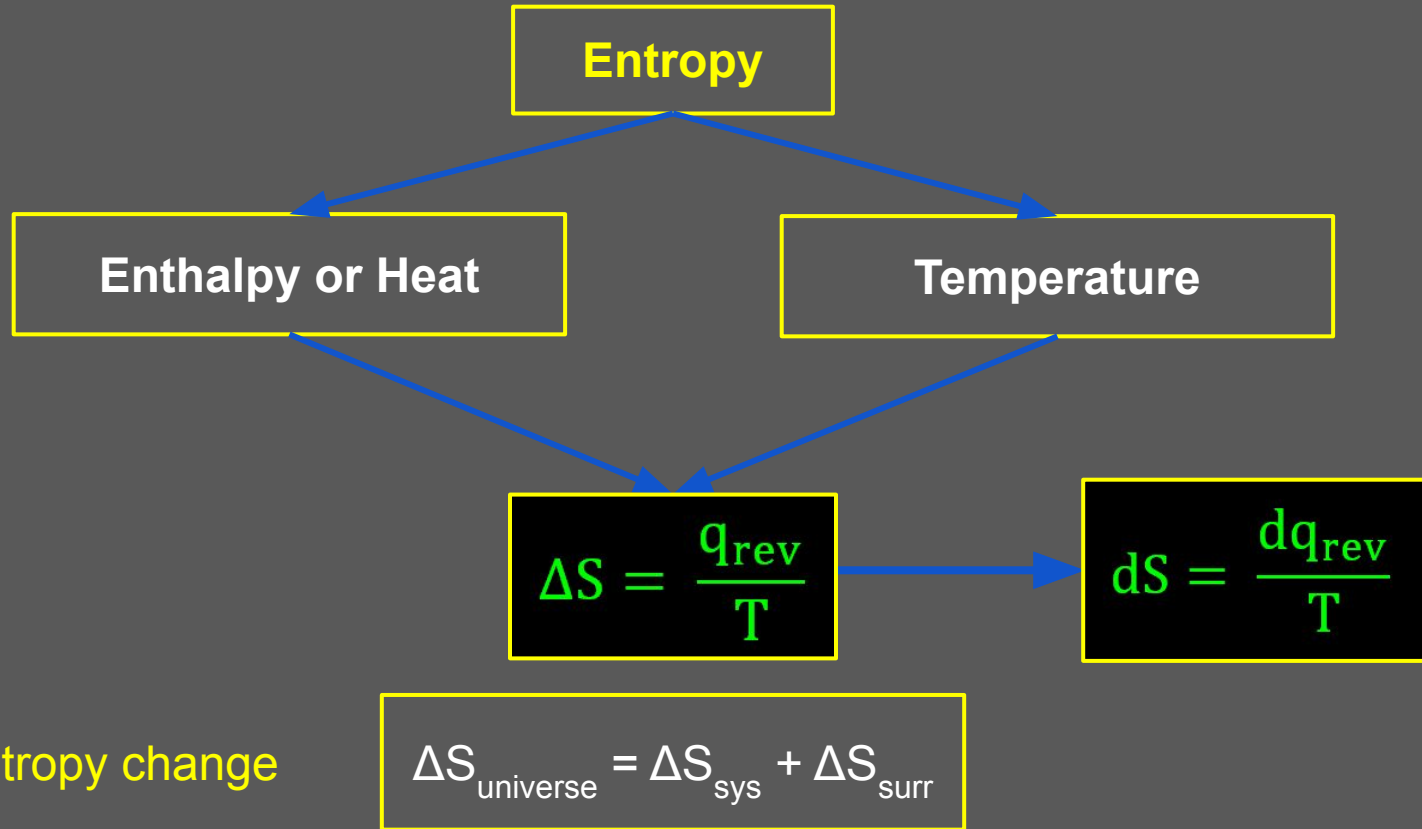
- i) A liquid crystallizes into solid
- ii) Temperature of crystalline solid raised from 0K to 115K
- iii) $2\text{NaHCO}_3(\text{s}) \longrightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$
- iv) $\text{H}_2(\text{g}) \longrightarrow 2\text{H}(\text{g})$

Sol.

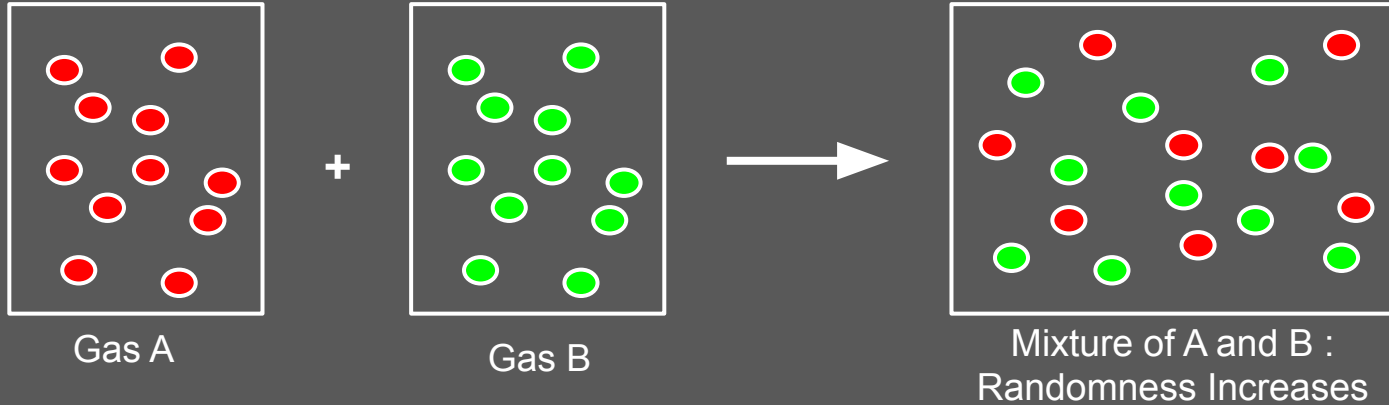
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- iv) $\text{H}_2(\text{g}) \longrightarrow 2\text{H}(\text{g})$

Sol.

- i) Entropy order solid << liquid < gas, so **entropy decreases**
- ii) Temperature raised, **more molecular motion, more entropy**
- iii) Product has **more gaseous molecules/atoms so increase in entropy**
- iv) Product has **more gaseous molecules/atoms so increase in entropy**



Entropy vs spontaneity (Second Law of Thermodynamics)



Second Law of
Thermodynamics

Entropy of universe
increases

$$\Delta S_{\text{universe}} > 0$$

11C06.5 Spontaneity

$$\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

ENTROPY FOR DIFFERENT PROCESSES

Reversible

$$\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T}$$

$$\Delta S_{\text{sys}} = \frac{q_{\text{sys}}}{T}$$

Under equilibrium, $q_{\text{sys}} = -q_{\text{surr}}$

Thus, $\Delta S_{\text{sys}} = -\Delta S_{\text{surr}}$

$$\Delta S_{\text{universe}} = 0$$

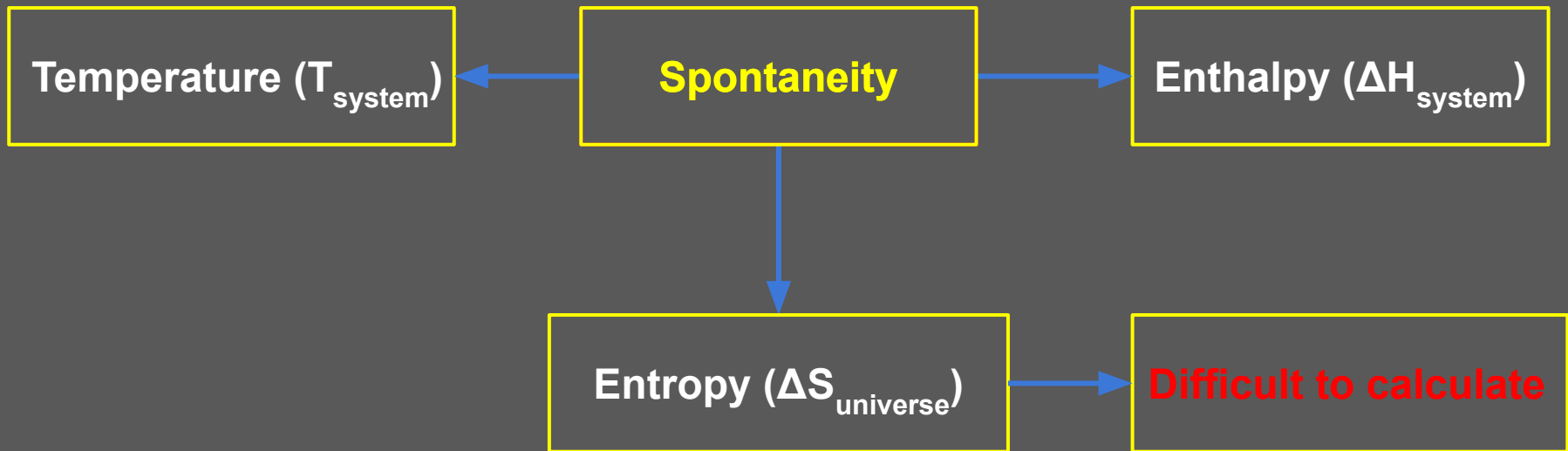
Irreversible

$$\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T}$$

as, $\Delta S_{\text{sys}} \neq -\Delta S_{\text{surr}}$

Since irreversible process
are spontaneous in nature

$$\Delta S_{\text{universe}} > 0$$



Property of System

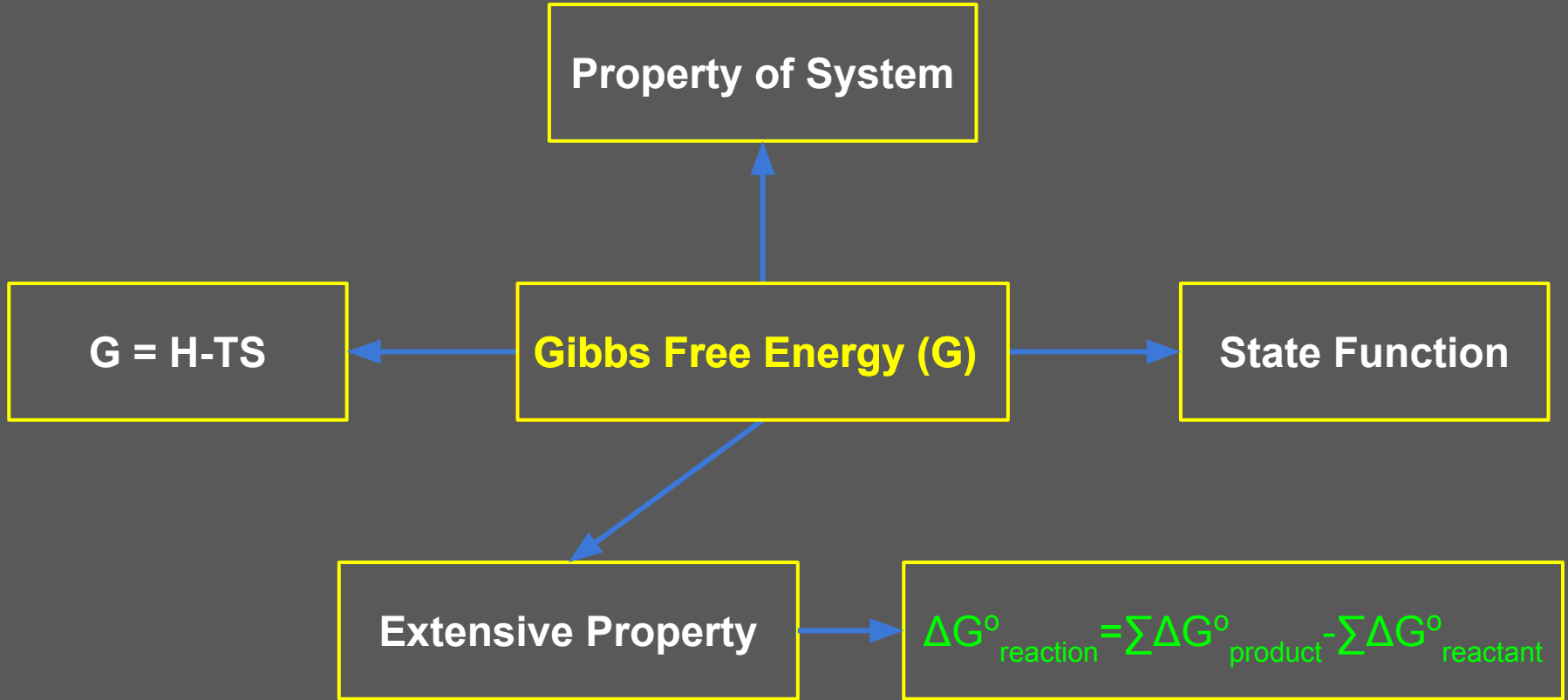
$$G = H - TS$$

Gibbs Free Energy (G)

State Function

Extensive Property

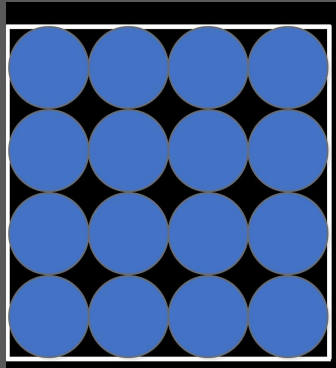
$$\Delta G^{\circ}_{\text{reaction}} = \sum \Delta G^{\circ}_{\text{product}} - \sum \Delta G^{\circ}_{\text{reactant}}$$



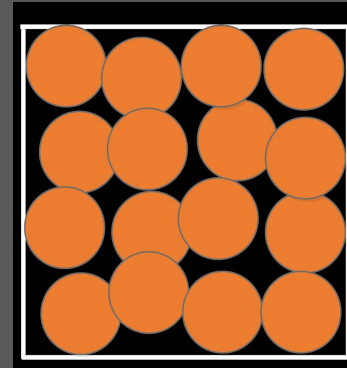
Third Law of thermodynamics



Entropy of pure crystalline substance at 0 K approaches ZERO



At 0 K no disorderness
so $S = 0$



Disorderness increases
so $S > 0$